

TECHNICAL SKILLS

Tech Stack	Python (Advanced), C++, Git, Bash, Flask/FastAPI, OpenCV, ROS2
MLOps & Cloud	CI/CD, VertexAI, Docker, Kubernetes, Google Cloud, AWS, DVC, Terraform
Data Science	Product Recommendation, Image Processing, A/B Testing, Feature Engineering
Data Engineering	SQL, BigQuery, Airflow, Pub/Sub, ETL, Data Acquisition, Sensor Fusion (LiDAR/Camera)
Communication	English (Fluent), Korean (Intermediate), Indonesian (Native)

PROFESSIONAL EXPERIENCE

Computer Vision Researcher

WicomAI Lab Kookmin University

Sep 2024 — Present

Seoul, South Korea

- Edge AI Deployment:** Fine-tuned and deployed YOLOv8 for real-time fire detection on **NPU-based CCTV with 81% accuracy**, leveraging **ONNX** for model quantization and engineering custom hardware-level post-processing pipelines in **C++**.
- Multimodal Segmentation:** Generated synthetic multimodal (RGB, IR) drone instance segmentation datasets utilizing **SAM** to train and deploy a lightweight, attention-based Encoder-Decoder model achieving **84% AP** on edge device.
- Performance Optimization:** Architected an optimized multiprocessing system on edge hardware to manage AI-based drone tracking, gimbal control, increasing system by image processing throughput **7x (30 FPS to 220 FPS)** for Drone OCC.
- Sensor Fusion Engineering:** Designed a ROS2-based multimodal (LiDAR, Camera) drone localization system, achieving high-precision depth estimation with **±5 cm error** at 20m range.
- Technical Leadership:** Spearheaded a research team of **5 graduate engineers**, defining technical roadmaps for Point Cloud processing, Panoptic Segmentation, and AI integration to assist Drone SLAM project.

MLOps Engineer

Hypefast (Largest House of Digital Native Brands in SEA)

Apr 2024 — Feb 2025

Jakarta, Indonesia

- MLOps Infrastructure:** Built the company's first end-to-end MLOps platform using **Google VertexAI, Docker, and DVC**. For automating retraining pipelines, experiment tracking, and reducing model deployment time.
- Data Platform:** Architected a data warehouse (data-lake, datamart) strategy separating raw ingestion from business logic layers for 11 distinct local brands, successfully reducing data availability from **24 hours to real-time**.
- High-Availability Integrations:** Developed containerized microservices on **Cloud Run and Docker** to unify Open API integrations across major marketplaces (Shopee, TikTok Shop, Lazada), achieving a **99.9% uptime SLA**.
- Pipeline Orchestration:** Engineered **20+ auto-scaling ETL pipelines** using **Airflow, BigQuery, and Pub/Sub** with built-in automated backfilling, schema validation, and real-time failure alerting mechanisms.

Data Scientist

Hypefast (Largest House of Digital Native Brands in SEA)

Mar 2022 — Mar 2024

Jakarta, Indonesia

- Revenue Impact:** Developed a hybrid recommendation engine (Collaborative + Content-based) serving 11 brands, directly driving **\$427k+ annualized revenue** and achieving a **3.6% Conversion Rate**.
- Experiment & Growth:** Executed **10+ A/B tests** comparing heuristic and ML performance for **Product Recommendation** and **Smart Voucher** systems across 11 Fashion and Health & Beauty brands, presenting strategic results directly to the Board of Directors through **Metabase**.
- Data Acquisition:** Developed an OCR extraction pipeline for shipping labels, successfully processing **500,000+ customer profiles** to fuel sales lead generation.
- Data Scraping:** Optimized Python scrapers to harvest competitor pricing data from major e-commerce platforms, handling **hundreds of thousands** of daily requests.

Data Scientist

PT PLN (Indonesia National Electricity Company)

Sep 2021 — Jun 2022

Jakarta, Indonesia

- Data Engineering:** Designed a scalable SQL database schema and Python script to ingest, clean, and standardize **millions of rows** of telemetry data from Jawa-Bali's power grid.
- Fraud Detection System:** Built end-to-end classification pipelines to identify electricity theft, achieving a **20% actual hit rate** in field inspections and recovering lost revenue.
- Interface Development:** Developed dashboard using **Streamlit** to visualize and analyze results of Battery Energy Storage System (BESS) algorithms.

EDUCATION

Kookmin University

Master of Science in Electrical Engineer (GPA: 4.43/4.5)

Sept 2024 — July 2026 (expected)

Seoul, South Korea

University of Indonesia

Bachelor of Engineering in Computer Engineer (GPA:3.72/4.0)

Aug 2018 — Aug 2022

Depok, Indonesia